

Project 1 – Temperature and Humidity monitoring using DHT22

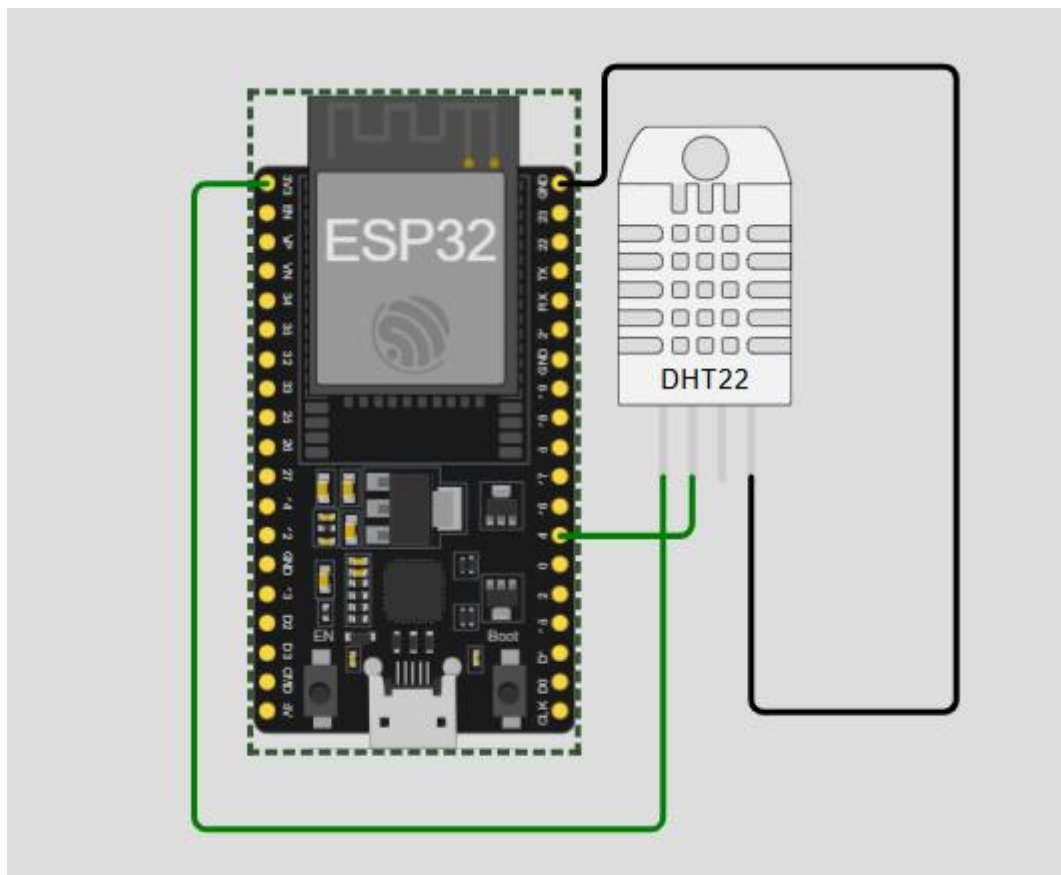
Objective:

The objective of this project is to measure environmental temperature and humidity using a DHT22 sensor connected to ESP32 and display the output in the serial monitor.

Components used:

- *ESP32*
- *DHT22 Sensor*
- *Jumper wires*
- *Wokwi simulator*

Circuit Diagram:



<i>Component</i>	<i>ESP32 Pin</i>
<i>DHT22 Data</i>	<i>GPIO 4</i>
<i>VCC</i>	<i>3.3V</i>
<i>GND</i>	<i>GND</i>

Working Principle:

- *DHT22 senses temperature and humidity*
- *ESP32 reads sensor values*
- *Values displayed on Serial Monitor*

Code:

```
#include "DHT.h"

#define DHTPIN 4
#define DHTTYPE DHT22

DHT dht(DHTPIN, DHTTYPE);

void setup() {
  Serial.begin(115200);
  dht.begin();

  Serial.println("-----");
  Serial.println(" Smart Environmental Monitor");
  Serial.println(" DecodeLabs - Project 1");
  Serial.println("-----");
}

void loop() {
  float humidity = dht.readHumidity();
  float temperature = dht.readTemperature();

  if (isnan(humidity) || isnan(temperature)) {
    Serial.println("Sensor read failed!");
    return;
  }

  Serial.println("==== Sensor Readings =====");

  Serial.print("Temperature : ");
  Serial.print(temperature);
  Serial.println(" °C");

  Serial.print("Humidity  : ");
  Serial.print(humidity);
  Serial.println(" %");

  Serial.println("=====");
  Serial.println();

  delay(2000);
}
```

Output:

```
-----  
Smart Environmental Monitor  
DecodeLabs - Project 1  
-----
```

```
==== Sensor Readings ====  
Temperature : 24.00 °C  
Humidity    : 40.00 %  
=====
```

Result:

The project was implemented successfully in Wokwi simulation and the desired output was obtained.

Conclusion:

Through this project, I learned sensor interfacing, GPIO pin configuration, simulation using Wokwi, and real – time data processing using ESP32.